

Samunder Singh

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Bengaluru, India

EDUCATION

- **Tezpur University** Tezpur, Assam
Bachelor of Technology in Electrical, Electronics and Communications Engineering 2020 – 2024

EXPERIENCE

- **CDAC Bangalore** Bengaluru, India
Design Engineer Jul 2026 – Present
 - **SLM Training & Inference:** Building end-to-end infrastructure for distributed SLM training and inference, covering training pipelines, inference optimization, and model quantization for in-house 1B–7B parameter models.
 - **Custom GPU Accelerator Stack:** Developing the software stack for a custom GPU accelerator at the compiler/runtime level using OpenCL kernels, a RISC-V POC, and POCL to run efficient AI/LLM workloads on the in-house GPU architecture.
 - **Frameworks & LLM Inference:** Enabling PyTorch-like AI frameworks and LLM inference on custom hardware, bridging low-level compiler and kernel development with high-level AI workloads across the hardware–software stack.
- **Lincode.ai** Bengaluru, India
Associate Data Scientist May 2024 – Jul 2026
 - **Synthetic Data Pipeline:** Built and shipped a pipeline with a structured defect library covering defect generation, augmentation, and dataset versioning to address rare-class imbalance in industrial inspection models.
 - **Distributed Training:** Set up training across multi-GPU, multi-host clusters using PyTorch DDP; integrated multi-bucket cloud storage (AWS S3, DigitalOcean Spaces, Storj) for dataset and artifact management.
 - **Defect Detection & Inference:** Developed and deployed computer vision models (YOLOv8, DETR, CLIP) for defect detection; optimized inference with torch.compile and TensorRT for edge and cloud serving.
 - **MCP Server & AI Tools:** Built an MCP server and custom AI tools that enabled LLM-driven automation of internal workflows.
 - **Clustering & Explainability:** Applied K-Means and DBSCAN for unsupervised defect pattern discovery; used Grad-CAM and SHAP to surface model explanations for non-technical stakeholders.

PROJECTS

- **Tvarant** (C++, PyTorch, OpenCL, CUDA): Out-of-tree PyTorch device backend with CPU simulation, an OpenCL runtime, and LLM-oriented JIT fusion.
- **SLX** (Slang, PyTorch, Python, CUDA): Custom deep learning framework on PyTorch using shader-based CUDA kernels for tighter kernel–op integration.
- **Distributed GPU Training & LLM Fine-Tuning** (PyTorch, DeepSpeed, Accelerate, Unsloth, TRL, k3s): Multi-node GPU environment (RTX 2060, AMD MI300X) with torchrun, NCCL/RCCL, k3s, and DeepSpeed ZeRO; fine-tuned Qwen3.5-9B with LoRA via Unsloth and HuggingFace TRL on Yelp review data.
- **YOLO LangChain Agent** (YOLO, LangChain, Python, OpenAI): Agentic system to train, validate, and export YOLO models via natural language commands.

OPEN SOURCE

- **Keras** (Aug 2024 – Present): 6 PRs implementing NumPy-compatible functions for the OpenVINO backend.
- **Unify.AI (Ivy)** (Jun 2023 – Present): 80+ PRs across JAX, PyTorch, TensorFlow, and PaddlePaddle; implemented activation and statistical functions for cross-framework compatibility.
- **CrossGL / CrossTL** (Jun 2024 – May 2025): Implemented a universal shader IR enabling bidirectional translation between shader languages.

PROGRAMMING SKILLS

- **Languages:** Python, C++, CUDA, SQL
- **ML & Frameworks:** PyTorch, TensorFlow, Keras, JAX, Scikit-learn, XGBoost, YOLOv8, DETR, CLIP, LSTM, CNN
- **Computer Vision:** OpenCV, Mediapipe, PaddleOCR, Grad-CAM, SHAP, TensorRT, torch.compile
- **Distributed Systems:** PyTorch DDP, DeepSpeed ZeRO, HuggingFace Accelerate, Unsloth, TRL, k3s, multi-GPU/multi-host training, edge deployment, inference optimization
- **Compilers & Hardware:** OpenCL, POCL, RISC-V, custom GPU runtime and kernel development
- **Backend & Cloud:** Python backends, MCP server, REST APIs, GCP, DigitalOcean, AWS S3, Storj, Docker, Apache Spark, DVC, ETL Pipelines
- **Data & Viz:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit